

NEW HILL (COLINA NOUĂ)



Why is this project a best practice?

Challenge

The project addresses the difficulty of scaling low-energy and healthy-building principles beyond a single dwelling. It had to combine density or phased delivery with reliable energy performance, good indoor conditions, landscape quality, mobility and construction cost, while keeping the sustainability measures practical for many future occupants.

Solution

The response combines a 140-hectare masterplan reported to preserve about 70% as green area; five green-house models supported by an international school, organic farm, café, recreation and planned co-working; and turnkey homes with contemporary energy-efficiency and smart-home systems in a rural landscape near Cluj. These measures were brought together through an integrated design approach and, where applicable, the Green Homes, nZEB or Passive House performance framework.

Replicability

The approach can be scaled through repeatable specifications for envelope, MEP systems, materials, commissioning, mobility and landscape. Replication is strongest when the developer applies the same measurable requirements to every phase and verifies them at handover, rather than treating certification as a one-off design exercise.

Key facts and figures

Location	Băbuțiu, Vultureni commune, Cluj County, Romania
Building use / function	Low-density residential community with education, farming, work and leisure facilities
Project type (new build, refurbishment, extension)	New build; phased neighbourhood development
Plot area / Floor area	Masterplan approximately 140 ha; first housing phase reported as 26 single-family buildings

Year / phase	Phased development; Green Homes-certified first phase
Climate zone	Humid continental; cool winters and mild-to-warm summers
Thematic focus / innovation	Green neighbourhood; 70% green areas; efficient houses; local food; school; work-from-home and community amenities
Investor / building owner	Colina Nouă
Architect	Arhimar is identified for housing design; international design specialists also involved
Other involved stakeholders	Romania Green Building Council; Colina Learning Center; Colina Farms; local contractors and community operators
Contact person	Colina Nouă - https://colinanoua.ro/en/
Further information	https://green-homes.org/property/colina-noua-1/ https://www.rogbc.org/proiecte-certificate-ro gbc?lang=en https://colinanoua.ro/en/

New Hill (Colina Nouă) is a new build; phased neighbourhood development project in Băbuțiu, Vultureni commune, Cluj County, Romania. It functions as low-density residential community with education, farming, work and leisure facilities. Its main best-practice contribution is Extends the best-practice discussion from individual buildings to a mixed community that connects housing, education, food and wellbeing. The documented strategy brings together a 140-hectare masterplan reported to preserve about 70% as green area, five green-house models supported by an international school, organic farm, café, recreation and planned co-working, and turnkey homes with contemporary energy-efficiency and smart-home systems in a rural landscape near Cluj. The project is phased; current status, available units and later masterplan stages should be verified directly with the developer. Public-source research was reviewed in July 2026; data that vary by phase are explicitly flagged in this sheet.

What was done differently?

1. Performance concept - A 140-hectare masterplan reported to preserve about 70% as green area.
2. Integrated systems and operation - Five green-house models supported by an international school, organic farm, café, recreation and planned co-working.
3. Wider value - Turnkey homes with contemporary energy-efficiency and smart-home systems in a rural landscape near cluj.

What was achieved?

Impact

Environmental impact: The project reduces energy and associated emissions primarily by lowering demand and then using efficient or renewable systems. Its material, landscape, waste or mobility measures add benefits beyond operational energy where documented. A verified whole-life carbon assessment was not publicly available, so the impact is described qualitatively unless a source provides a specific figure.

Economic impact: Lower energy demand can reduce utility exposure and improve long-term operating-cost predictability. Standardised high-performance specifications and third-party certification can also support quality assurance, market value and access to green-

finance products. Public sources do not provide a complete investment, payback or life-cycle-cost analysis for this project.

Social impact: The main social benefits are stable indoor temperatures, better air quality, daylight and acoustic comfort, together with safer and more attractive shared or outdoor spaces where these form part of the project. The project also raises market awareness by making high-performance housing visible to residents, designers, contractors and investors.

Key Learnings

Success factors: Key success factors are an integrated design process, a clear performance framework, early coordination of envelope and building services, selection of competent suppliers, quality control during construction and commissioning. Transparent documentation and communication with occupants are essential to preserve the intended performance in use.

Lessons for replication: Replication should start with an energy and comfort target, not a catalogue of technologies. Design teams should verify thermal bridges, airtightness, ventilation, controls and renewable sizing together; record assumptions by project phase; commission systems; and monitor early operation. Where public figures differ by phase, the final as-built dataset must become the single source of truth.